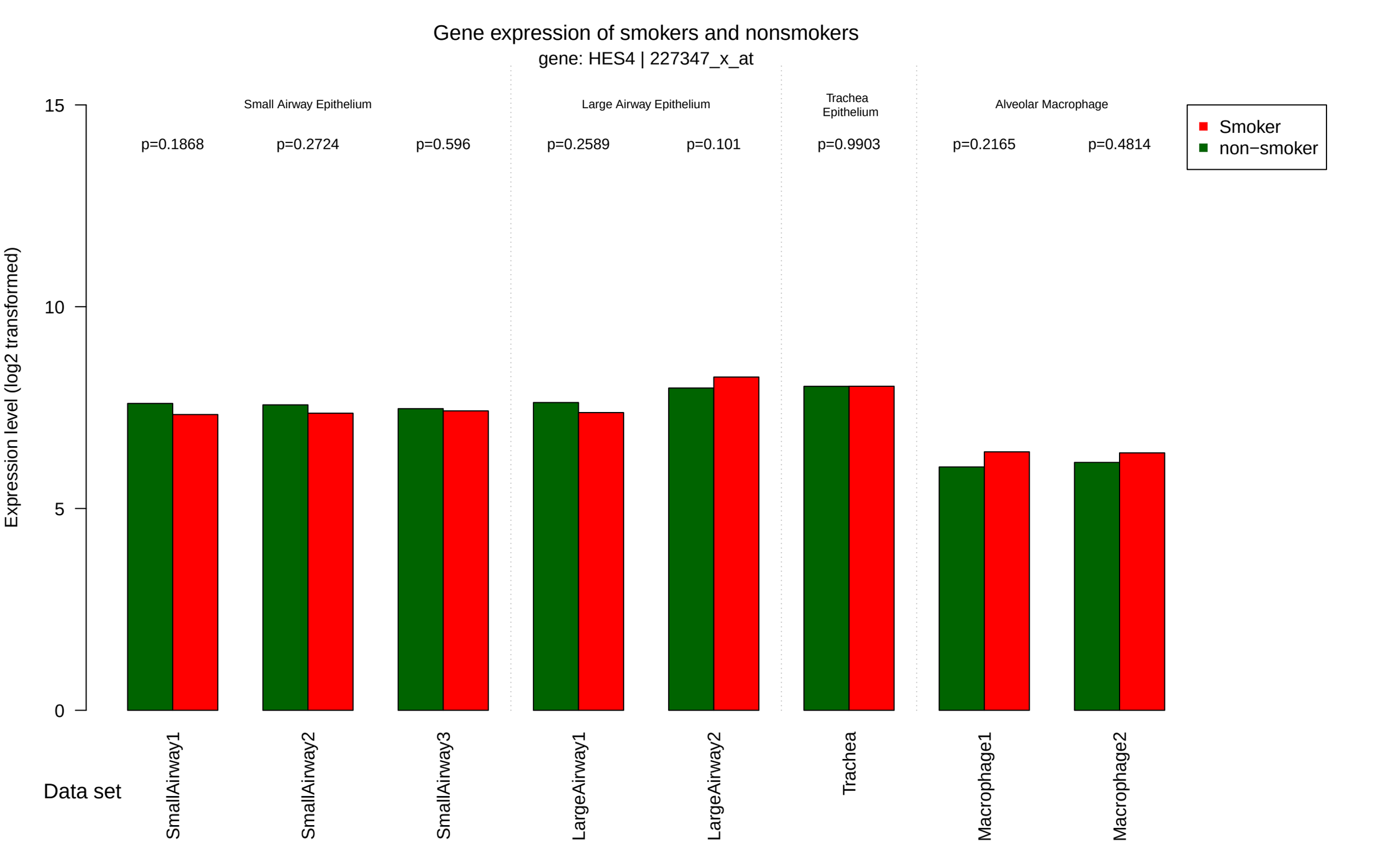
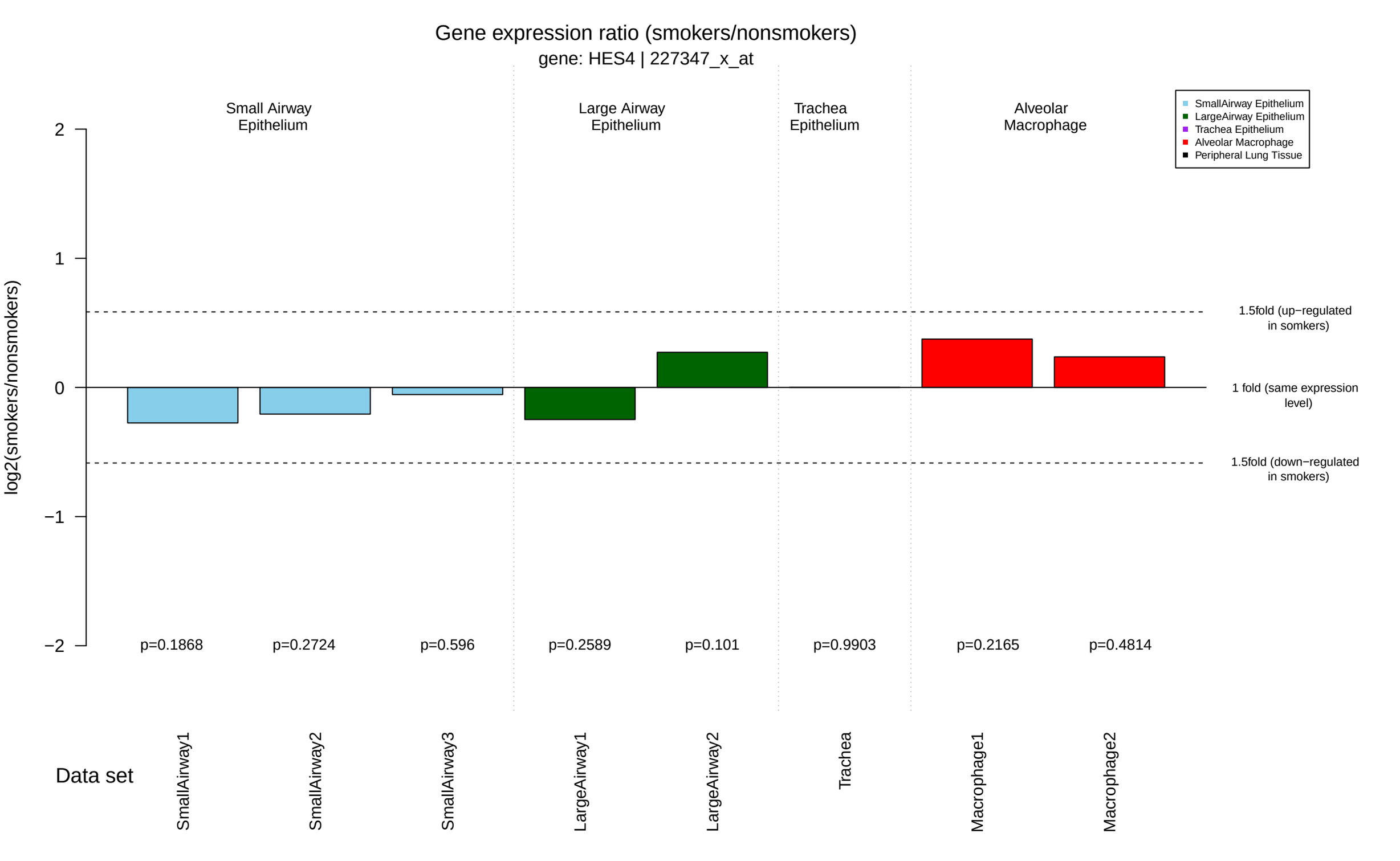
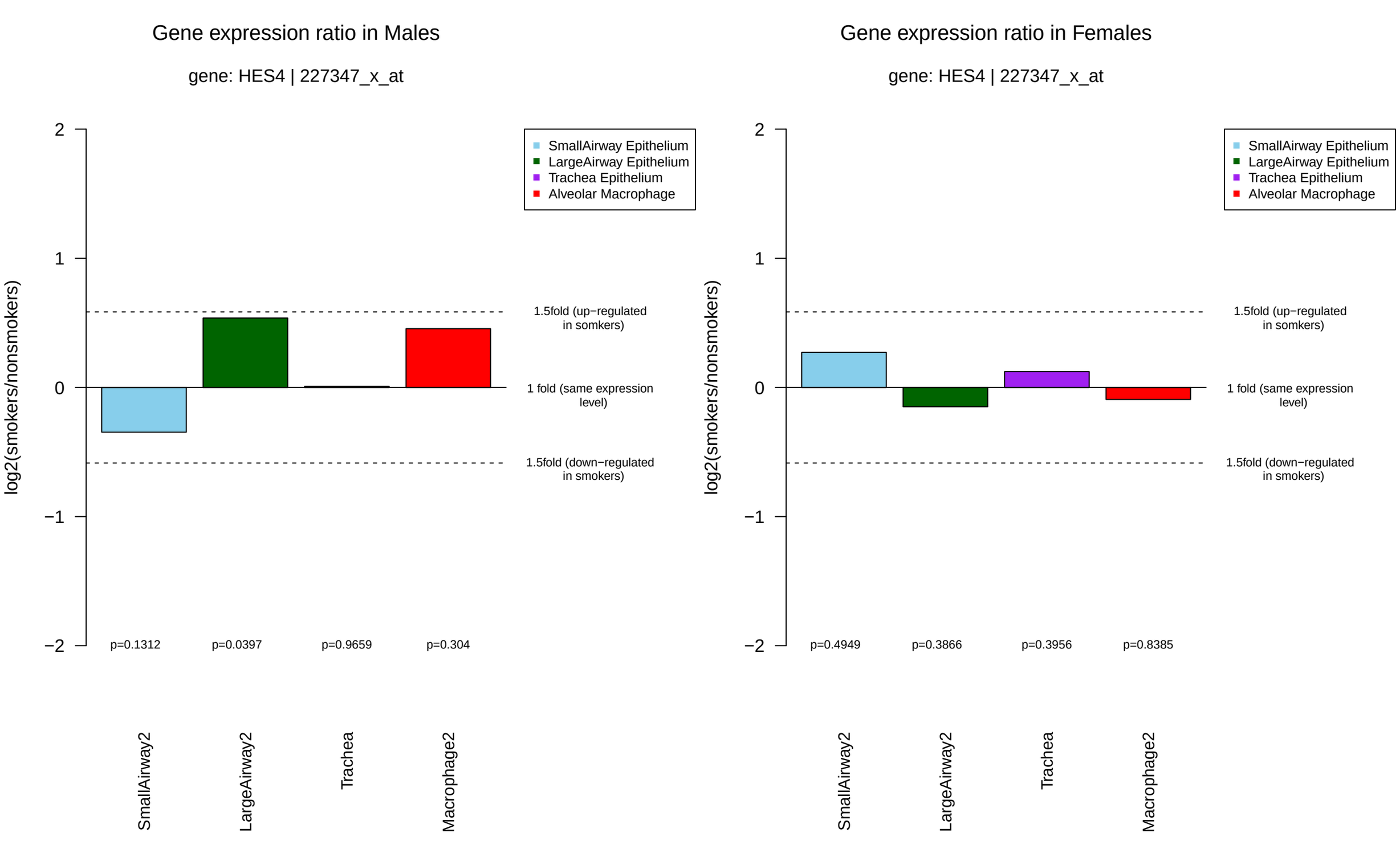
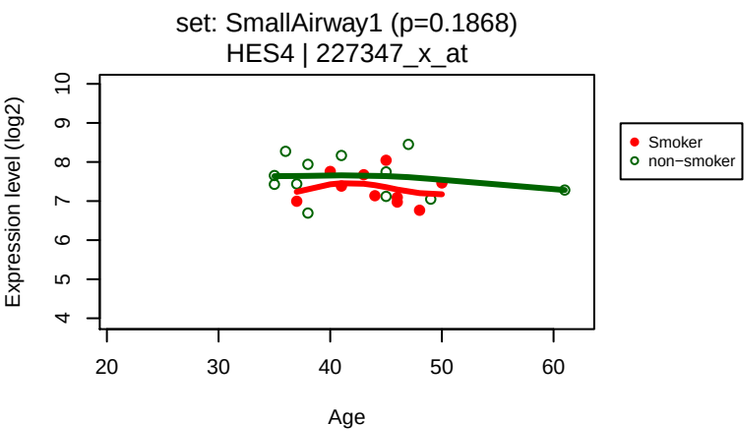
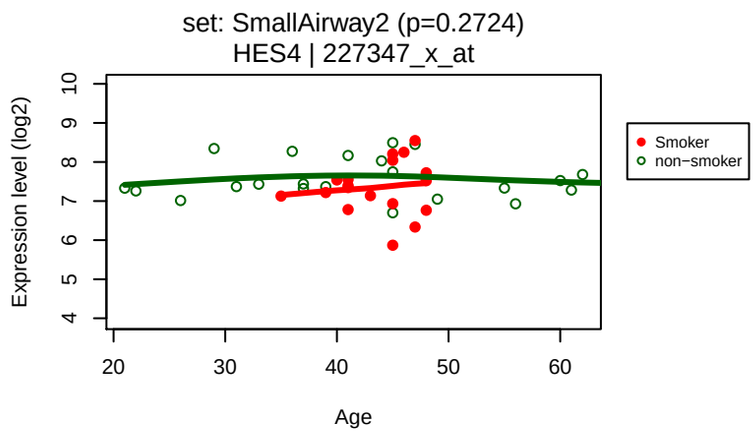
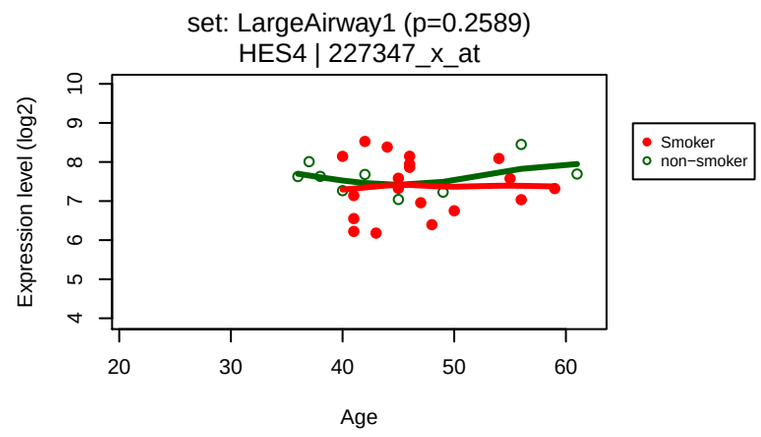

Gene expression of smokers and nonsmokers
gene: HES4 | 227347_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1868
p=0.2724
p=0.596
p=0.2589
p=0.101
p=0.9903
p=0.2165
p=0.4814
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

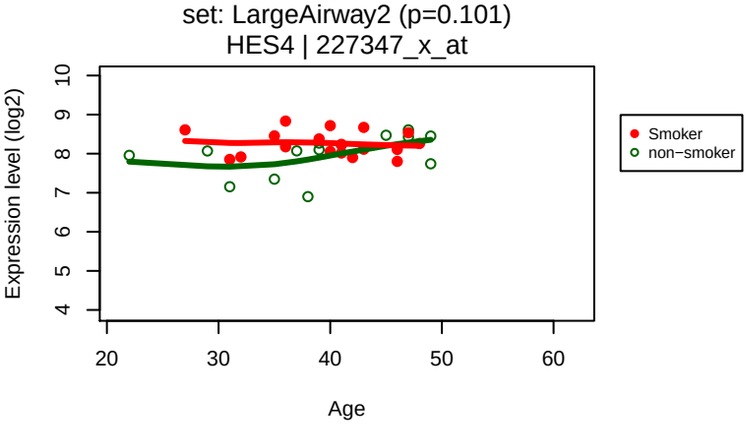

Gene expression ratio (smokers/nonsmokers)
gene: HES4 | 227347_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1868
p=0.2724
p=0.596
p=0.2589
p=0.101
p=0.9903
p=0.2165
p=0.4814
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

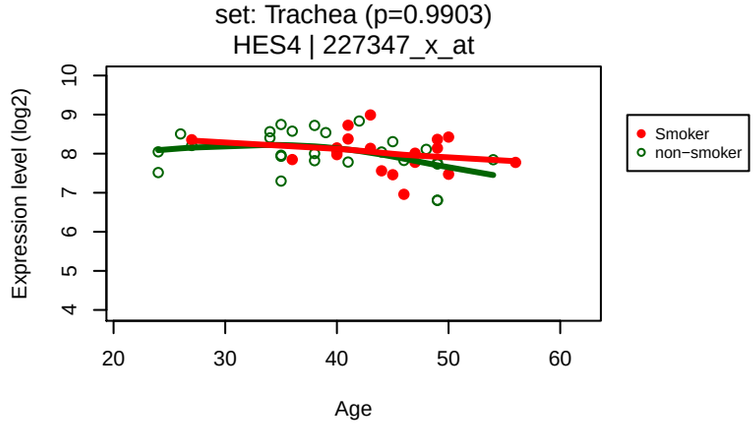

Gene expression ratio in Males
gene: HES4 | 227347_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1312
p=0.0397
p=0.9659
p=0.304
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HES4 | 227347_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4949
p=0.3866
p=0.3956
p=0.8385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

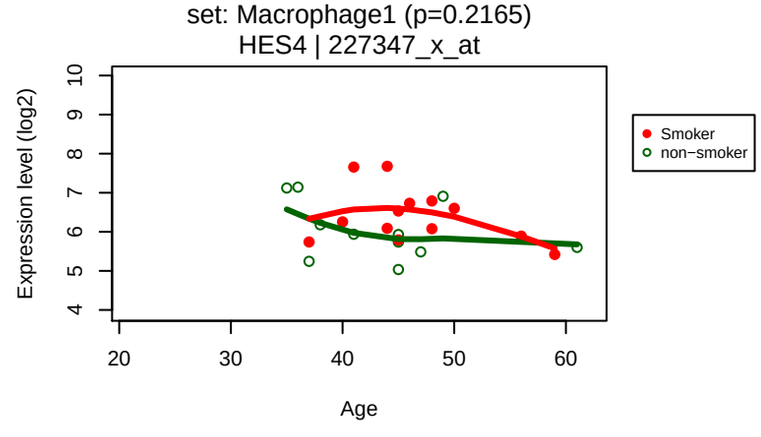

set: SmallAirway1 (p=0.1868)
HES4 | 227347_x_at
Expression level (log2)
Age
Smoker
non−smoker

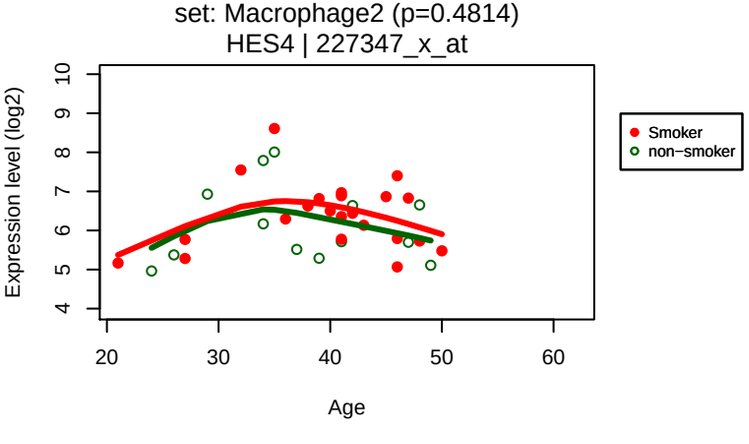

set: SmallAirway2 (p=0.2724)
HES4 | 227347_x_at
Expression level (log2)
Age
Smoker
non−smoker

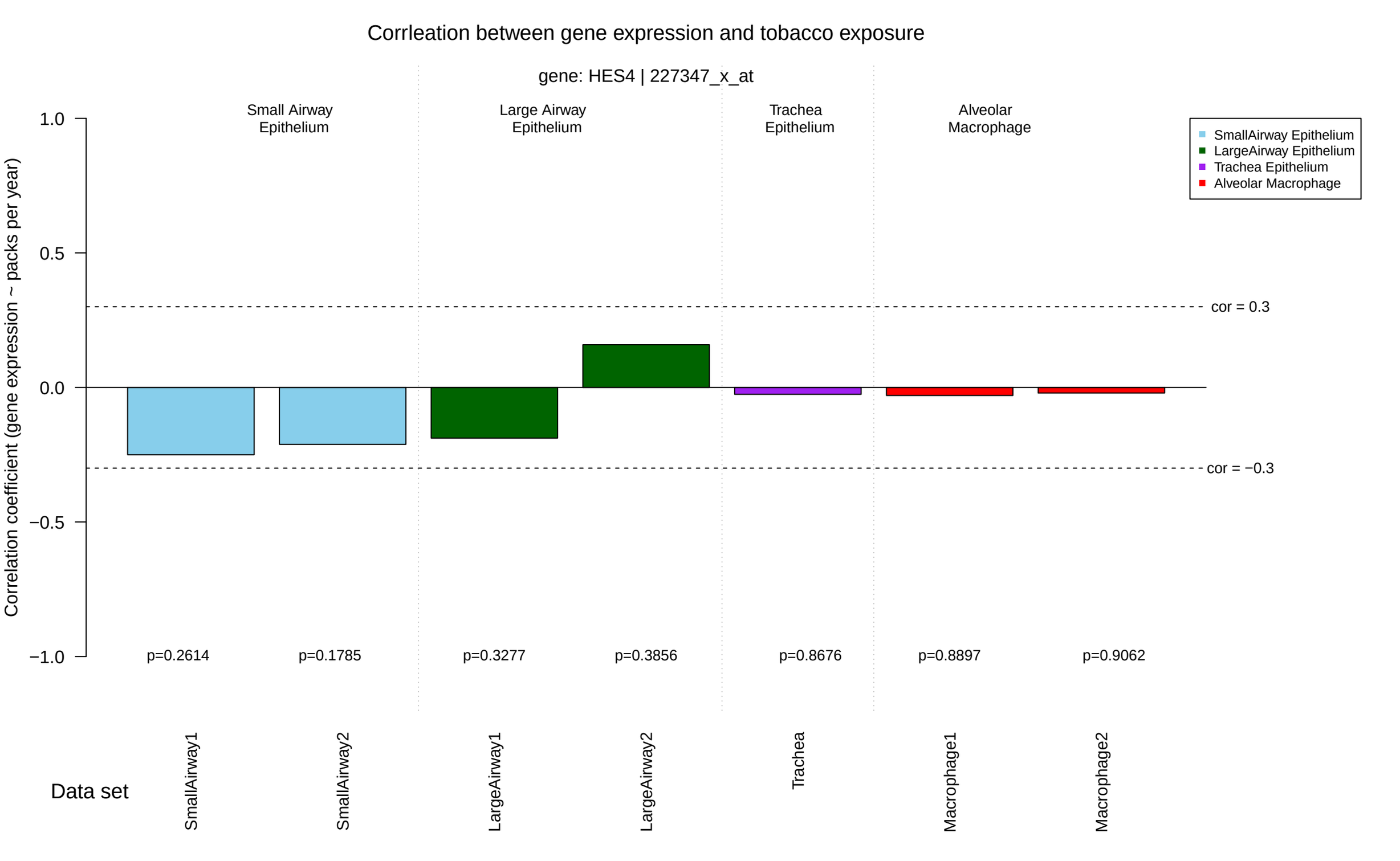

set: LargeAirway1 (p=0.2589)
HES4 | 227347_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.101)
HES4 | 227347_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9903)
HES4 | 227347_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2165)
HES4 | 227347_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4814)
HES4 | 227347_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HES4 | 227347_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2614
p=0.1785
p=0.3277
p=0.3856
p=0.8676
p=0.8897
p=0.9062
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2